on time Complexity and methods

A. Pogitha
192511230

→ Compare time Complexity of all notation in bubble, insertion, selection, merge, quick, heap

→ Name	Best	Average	Worst
Bubble	n	n^2	n^2
Insertion	n	n^2	n^2
~~Shell~~ Selection sort	n^2	n^2	n^2
~~Heapsort~~ Shell	$n(\log_3 n)$	$n^{3/2}$	$n^{3/2}$
Heap	n	nlogn	nlogn
Quick	nlogn	nlogn	n^2
merge	nlogn	nlogn	nlogn

Feature	hashing	binary search
Basic Idea	uses a hash function to map a key to an index O(1)	O(logn) Repeatedly divides a sorted array into two halves
Data requirement	Does not require ~~to~~ sorted data	Data must be sorted
Best case	O(1)	O(1)
Average case	O(1)	O(logn)
Worst case	O(n) due to collision	O(logn)

Search method	Hash function + hash table	Compare with middle element
Insertion	Average O(1)	O(n) in an array if maintaining sorted
Deletion	Avg O(1)	O(n) in an array
Main issue	Collision handling	Requires sorted data